Status	Finished
Started	Tuesday, 20 October 2026, 3:00 PM
Completed	Tuesday, 20 October 2026, 3:06 PM
Duration	6 mins 30 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort

b. Selection sort

c. Heap sort

d. Merge sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort

b. Binary Search

c. Linear Search

d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Insertion sort

d. Bubble sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists

b. Sorting the entire list sequentially

c. Comparing each element with the others

d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort

b. Bubble Sort

c. Merge Sort

d. Quick Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Sorting
- c. Merging
- d. Searching

The correct answer is:
Complexity

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Searching
- c. Merging
- d. Sorting

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It requires external libraries
- c. It is less efficient than custom sorting algorithms
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By sorting data in a single pass
- b. By comparing elements sequentially
- c. By using the bubble-up method
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. arrange()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Deleting data from a list
- c. Inserting data into a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are discarded
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **14**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Greedy method
- b. Dynamic programming
- c. Backtracking
- d. Divide and conquer

The correct answer is:
Divide and conquer

Question **15**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Sorting data in a single pass
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions